

Phase 1 & Phase 2 Solution Plan

What we build, requirements, algorithms, and how to demo to external examiners

NeuroShift uses camera context and sparse EMG to decide which home device a motor-impaired user intends to control - and abstains when unsure - to reduce false actuations.

1. Our solution

Camera finds feasible devices. EMG confirms user effort. Intention policy chooses ACT or ABSTAIN. ESP32+MQTT controls lamp/fan/plug. Dashboard logs why.

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Camera -> Objects + Pose -> Candidate devices (affordances)
EMG    -> Volition score    -> Is user trying to act?
Context-> Habit prior (P2)  -> Reweight options
      |
      Intention Policy -> Act / Abstain
      |
      MQTT + ESP32 -> Lamp / Fan / Plug
  
```

2. Phase split

	Phase 1 (Sem 7 NOW)	Phase 2 (Sem 8 NEXT)
Goal	Working prototype + pilot data	Research proof + IEEE paper
Fusion	Rule-based Act/Abstain	Learned fusion + calibrated abstention
Context	Device state + basic time	Personal habit prior
XAI	Decision log	Caregiver explanations + dashboard
Demo	Live hardware demo	Live demo + graphs + paper
Deliverable	Prototype + report + dataset start	Evaluation + paper + MVP polish

3. Phase 1 - What we build

- EMG acquisition (MyoWare -> ESP32/Python)
- Volition encoder: P(user trying to act)
- Vision: YOLO/RT-DETR + MediaPipe/RTMPose
- Affordance map: objects -> candidate device actions
- Basic intention policy: Act or Abstain
- MQTT -> ESP32 relay / smart plug
- Logger + simple React dashboard

Phase 1 algorithms

Stage	Algorithm	Why
EMG preprocess	RMS / MAV features	Standard for sparse EMG
Volition	Threshold or small 1D-CNN/TCN	Effort detection, not gesture race
Objects	YOLOv8 or RT-DETR	Real-time; same family as base paper
Pose/hand	MediaPipe or RTMPose	Fast on laptop webcam
Affordance	Rule table object->action	Explainable in viva
Fusion	Late fusion rules + thresholds	Extends Chamberland gating
IoT	MQTT + ESP32 relay	Reliable live demo

Phase 1 fusion logic

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IF EMG_volition < T_low: ABSTAIN
```

NeuroShift 2.0 | Phase 1 & Phase 2 Solution Plan

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ELSE IF one feasible device AND score > T_act: ACT  
ELSE IF multiple feasible devices: ABSTAIN (or top if dominant)  
ELSE: ABSTAIN
```

4. Phase 1 requirements

Hardware (must have)

- MyoWare EMG + electrodes
- ESP32 DevKit
- Relay module (2-3 channels) or MQTT smart plug
- USB webcam + laptop
- Lamp + fan (demo devices)
- Mosquitto MQTT broker, wires, power supply

Software

- Python, PyTorch, OpenCV, YOLO/RT-DETR, MediaPipe
- FastAPI + React + SQLite/PostgreSQL
- ESP32 firmware + paho-mqtt

5. Phase 2 - What we add

Module	Algorithm / approach
Learned fusion	MLP / small cross-attention over EMG+vision+context
Calibrated abstention	Reject option / selective prediction tuned for FAR
Personal prior	Time-of-day + recent device-use counts/sequence model
Decision XAI	Structured reason: objects, EMG, prior, decision
Caregiver dashboard	React history + alerts + explanations
Evaluation	FAR, F1, ablations vs 4 baselines
Paper	IEEE EMBC-style draft

6. How to DEMO to external (8-10 min)

Time	Show	Say
0-1 min	Need slide with population data	~100M stroke survivors; India ~9.4M
1-2 min	Base paper vs NeuroShift	Gestures -> home intention + abstain
2-3 min	Architecture	Sensors -> policy -> MQTT -> devices
3-6.5 min	LIVE DEMO (3 scenes)	See scenes below
6.5-8 min	Results table	FAR vs baselines
8-9 min	Sem7 done / Sem8 next	Clear roadmap
9-10 min	Q&A	Novelty, ethics, limits

Live demo scenes (must rehearse)

- Scene A CLEAR: hand/lamp focus + EMG effort -> lamp ON (success)
- Scene B AMBIGUOUS: lamp+fan visible, weak signal -> ABSTAIN (safety)
- Scene C CONTRAST: always-act/EMG-only fails; NeuroShift abstains
- Scene D (optional): normal hand movement without intent -> ABSTAIN

Demo survival tips

- Pre-wire and tape relays night before
- Keep backup video if live fails
- Fix camera position; high-contrast objects
- Project dashboard decision panel
- Never skip the abstain scene - that is your novelty

7. External examiner Q&A cheat sheet

They ask	You answer
Novelty?	Home intention + Act/Abstain, not gesture accuracy
Base paper?	Chamberland EMBC 2024; extend gesture rescue to home intention
Why useful?	Stroke/paralysis population + independent living
Algorithms?	YOLO/MediaPipe + EMG volition + fusion policy + MQTT
Evaluation?	FAR + intention F1 vs EMG-only / vision-only / Chamberland-gate
If it fails?	Occlusion/noise -> abstain by design
Ethics?	Consent for data; not a clinical medical device claim in Phase 1

8. Phase 1 week plan

Weeks	Focus
1-2	MyoWare + ESP32 stream + relay blink
3-4	YOLO + MediaPipe + affordance map
5-6	Fusion rules + MQTT end-to-end control
7-8	Dashboard + logger + pilot data collection
9-10	Baselines + FAR table + polish demo
11-12	Report, PPT, external rehearsal

9. Do NOT do

- Chase Ninapro SOTA / Transformer as novelty
- Add fall/medicine/wheelchair into Phase 1 claims
- Depend on cloud LLM for core decision
- Demo without an abstain scene

10. Ownership sentences

Phase 1: We built a working EMG + vision + IoT intention prototype with Act/Abstain and pilot data.

Phase 2: We proved affordance-conditioned fusion + abstention + personal prior reduces false actuations vs baselines, and wrote the IEEE paper.